



Towards dynamic safety assurance for Industry 4.0

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ABSTRACT

The goal of Industry 4.0 is to be faster, more efficient and more customer-centric, by enhancing the automation and digitalisation of production systems. Frequently, the production in Industry 4.0 is categorised as safety-critical, for example, due to the interactions between autonomous machines and hazardous substances that can result in human injury or death, damage to machines, property or the environment. In order to demonstrate the acceptable safety of production operations, safety cases are constructed to provide comprehensive, logical and defensible justification of the safety of a production system for a given application in a predefined operating environment. However, the construction and maintenance of safety cases in alignment with Industry 4.0 are challenging tasks. For their construction, besides the modular, dynamic and reconfigurable nature of Industry 4.0, the architectural levels of the things, fog and cloud computing have to be considered. The safety cases constructed at system design and development phases might be invalidated during production operations, thus necessitating some means for dynamic safety assurance. Moreover, flexible manufacturing in Industry 4.0 also underlines the need for safety assurance in a dynamic manner during the operational phase. Currently published studies are not explicitly supporting the safety assurance of Industry 4.0, which is the focus of this paper with special emphasis on dynamic safety assurance. At first, the Hazard and Operability (HAZOP) and Fault Tree Analysis (FTA) techniques are used for the identification and mitigation/elimination of potential hazards. Next, based on the hazard analysis results, we derived the safety requirements and safety contracts. Subsequently, safety cases are constructed using the OpenCert platform and safety contracts are associated with them to enable necessary changes during runtime. Finally, we use a simulations based approach to identify and resolve the deviations between the system understanding reflected in the safety cases and the current system operation. The dynamic safety assurance is demonstrated using a use case scenario of materials transportation and data flow in the Industry 4.0 context.

1. Introduction

The main enablers for the fourth industrial revolution (aka Industry 4.0) are enhanced automation and digitalisation, which tend to make manufacturing processes faster, more efficient and more customer-centric. However, the enhanced automation and digitalisation, if not carefully orchestrated, could potentially lead to production failures or mishaps, making them safety-critical. Any industry is categorised as safety-critical if an unplanned event or sequence of events can potentially harm humans (injuries or even deaths) or create damages to machines, property or the environment. The principal objective of system safety is the identification, prevention, mitigation, and documentation of potential hazards. It is a fundamental element and a

regulatory requirement for safety-critical systems. The hazard elimination or mitigation is much more cost effective during system design and development phase than trying to inject safety after the occurrence of an accident or mishap [1].

To perform the safety analysis for Industry 4.0, besides the modular, dynamic and reconfigurable nature, the things, fog/edge and cloud levels need to be considered. Transformation to the Industry 4.0 significantly increases the safety assurance challenges [2]. To demonstrate the acceptable safety of Industry 4.0, the assurance (safety) cases are constructed that provide comprehensive, logical and defensible justification of the safety of a production system for a given application in a predefined operating environment [3]. In the context of Industry 4.0, however, the parts of safety cases constructed during system design and

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development phase may turn out to be incorrect, inapplicable or insufficient during the operation. This can be caused by emergent behaviours and changes performed in consequence of market demands, hazardous conditions or system failures. To exploit the full potential of Industry 4.0, there is a need to support the dynamic safety assurance [4,5].

There exist few studies on dynamic safety assurance. The research in [4] presents the idea of through-life safety assurance. It is reflected in four activities: identify, monitor, analyse and respond. Jaradat and Punnekkat [6] exploit safety contracts for monitoring the failure rates during operational phase. Calinescu et al. [7] develop the partial assurance argument at design phase to which the assurance evidence can be synthesised during self-adaptation. McDermid et al. [5] suggest the continued and proactive assessment of autonomous systems to address the mismatches between the ‘world as imagined’ system and its operational environment we considered and the ‘world as observed’ analysis based on the operational data. Jaradat et al. [2] highlight the safety assurance challenges for Industry 4.0. The published studies are not explicitly supporting the dynamic safety assurance by using the operational data.

This paper focuses on the dynamic safety assurance of Industry 4.0, where our specific contributions are: (a) proposing an overall approach incorporating Industry 4.0 specific aspects (b) elaborating on the details of individual steps including suggestions for an overall tool chain and (c) demonstrating the approach using a synthetic, but realistic use case scenario. We now provide an overview of the steps in our approach. First, we performed the hazard analysis, for which the Hazard and Operability (HAZOP) and Fault Tree Analysis (FTA) techniques are utilised for the identification and mitigation of potential hazards. Second, based on the hazard analysis results, we derived the safety requirements to prevent or mitigate the identified hazards. Safety contracts have also been derived for uncertainty sources that provide the means to detect deviations from intended behaviour and perform necessary adaptations at the operational phase. Third, we constructed the safety cases and associated the safety contracts with them. For this reason, the nature of Industry 4.0 and its generic architecture are explicitly taken into consideration. Finally, the operational data is utilised to resolve the deviations between the intended behaviour reflected in safety cases and the actual behaviour. In particular, based on the gathered data, the safety contracts constructed for uncertainty sources are monitored, deviations between the intended and actual behaviour are tracked and evaluated, and the safety contracts and safety cases are updated. The work presented in this paper utilises the OpenCert¹ platform that provides support for modelling and visualising the safety cases in Goal Structuring Notation (GSN) [3]. The usefulness of dynamic safety assurance is demonstrated for the flow of materials and data in an Industry 4.0 oriented use case scenario.

The rest of this paper is organised as follows: Section 2 describes essential background information on Industry 4.0, safety assurance and safety contracts. Section 3 presents the proposed methodology for dynamic safety assurance of Industry 4.0. Section 4 demonstrates the applicability of proposed methodology for materials transportation and data flow in Industry 4.0. Section 5 discusses the related work. Section 6 concludes the paper and presents future research directions.

2. Background

2.1. Industry 4.0

The association of production with the information and communication technologies lead to the Industry 4.0. It is a successor of three industrial revolutions. The first revolution was triggered by the steam and hydroelectric power generation; the machines were developed for the manufacturing of products. The second revolution was driven by

the electrification, which paved the way towards the mass production and assembly lines. The third revolution focused on the electronics and information technology; the programmed machines such as robots and Automated Guided Vehicles (AGVs) improved the automation. The fourth revolution is based on the four integral pillars: interoperability, information transparency, technical assistance and decentralised decisions [8]. Interoperability is the capability to exchange data between machines that enables joint collaboration; Internet-of-Things (IoT) is exploited for interconnecting machines, the ultimate intention is whole automation. A novel enabler of information transparency could be through the concept of digital twins [9]: leverage the virtual equivalents such as simulation models for assessing and improving the real products. Technical assistance provides the means to understand and control events that needs to be supported by systems. In this context, the comprehensive visualisations facilitate human operators to solve the problems in short time frames [10]. The intention of decentralised decisions is to incorporate automatic decision making capabilities in machines so that the human intervention in manufacturing process is not required, which makes the operator rather a supervisor.

The manufacturing process is re-defined for Industry 4.0. Its generic architecture consists of three levels: things, fog/edge and cloud. The things and fog/edge represent the local part of the system, whereas the cloud is a remote infrastructure that is typically owned by a third-party service provider. The things interact with the physical environment via different sensing/actuating devices. However, due to their limited storage and processing power, devices from things level rely on the fog or cloud for storage and processing services. The fog acts as a bridge between the things and the cloud. It receives data from the things and perform partial processing, especially for critical functions, or otherwise forward the data to the cloud infrastructure. Likewise, the fog directly instruct to the things, but the cloud forwards commands to the things via fog servers.

2.2. Safety assurance

An assurance (safety) case provides the evidence(s) to support the safety claims such as system behaviour and hazards addressed, and shows why the included evidence is adequate. In particular, the safety cases facilitate the knowledge exchange between different stakeholders, for example, suppliers and acquirers, and between operator and regulator [11]. To document safety cases, several approaches exist, such as free text, tabular structures and graphical notations [12]. Structured Assurance Case Metamodel (SACM) [11] is the Object Management Group (OMG) standard that integrates and standardises the broadly used notations for documenting safety cases including GSN. The work presented in this paper utilises the OpenCert platform that provides support for modelling and visualising the safety cases in GSN. Common Assurance and Certification Metamodel (CACM) implemented in OpenCert internally uses the SACM metamodel. The OpenCert platform also includes a Connected Data Objects (CDO)² server, which is a development-time model repository as well as a run-time persistence framework. It stores the safety case (i.e., model and diagram) in a database such that different distributed stakeholders can access them and work on the same safety case concurrently.

The main purpose of GSN is to show how goals (claims about the system behaviour, safety requirement or process plan), are broken down into sub-goals (sub-claims) and supported by available solutions (evidences). The rationale for decomposing the goals into sub-goals is documented by strategy (argument reasoning), whereas the scope and domain in which claim or strategy should be interpreted is done in the context element. Assumption element is an intentionally unsubstantiated statement, which is assumed to be true for a certain goal or strategy. Justification element provides an explanation or rationale

¹ <https://www.polarsys.org/opencert/>.

² <http://www.eclipse.org/cdo/>.

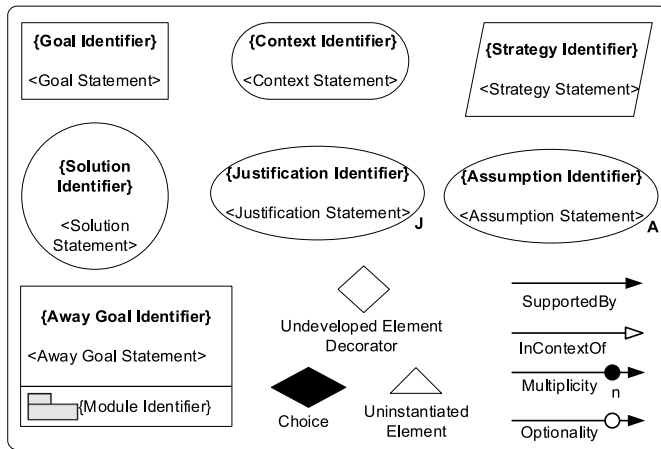


Fig. 1. GSN notations.

why a certain goal or strategy is considered appropriate or acceptable to be used [3]. The argument elements can be linked with one of the two relationships: SupportedBy and InContextOf. The SupportedBy relationship is used to show the inferential or evidential relationships between goals, strategies and solutions elements, whereas InContextOf relationship is used to declare contextual relationships of goals or strategy with context, assumption and justification elements. An away goal referenced to a claim presented in another argument module.

The principal elements of the GSN are shown in Fig. 1, for further details we refer the readers to GSN standard [3]. In order to represent patterns of argument, GSN has been extended to support structural abstraction aspects. Multiplicity relationship shows the n-ary (zero or more) relationships between elements, whereas optionality relationship denotes the alternative (zero or one) relationships between elements. Undeveloped and uninstantiated decorators are applied to the elements that need to be further developed and to be replaced (instantiated) with a more concrete instance, respectively. A choice element is used to denote possible alternatives (1-of-n and m-of-n selection) in satisfying a relationship.

2.3. Safety contracts

The contracts can be used almost everywhere and all phases of system life cycle, for instance, from early requirements to detailed design, for managing the changes in the system or its corresponding assurance case. Contracts were proposed by Bertrand Meyer as the concept of ‘design-by-contract’ for verification of software program [13]. In particular, the correctness of requirements is described as a contract between a method and its callers. The contracts are made up of two parts: (1) the preconditions that must be true while the initiation of program; and (2) the postconditions that must be true on program completion. A contract can be seen as a pair of assertions $C = \langle A, G \rangle$ i.e. {Assumptions, Guarantees} that describes in a formal way, under which context the design is assumed to operate, and what are its obligations [14]. By using the contracts, the behaviour of the component can be described in a way that the component makes assumptions (conditions) on its environment and if those assumptions A hold then the component will behave as guaranteed G (offers properties). The safety contracts can be associated with the safety case for supporting the dynamic assurance. They provide a mechanism for recording and justifying the agreed relationship between safety case modules that might be illustrated as a pattern in GSN.

3. Dynamic safety assurance of industry 4.0

The intention of the safety assurance of Industry 4.0 is to provide confidence that the risk of potential mishaps and accidents is eliminated or otherwise mitigated to an acceptable level. For this reason, the assurance (safety) case is constructed, in particular, a logical and appealing argument supported by a body of evidence, to justify that an Industry 4.0 oriented application will operate as intended in a defined environment. In the assurance case of Industry 4.0, besides the modular, dynamic and reconfigurable nature, the things, fog and cloud computing levels need to be reflected. Due to the enhanced automation and digitalisation of manufacturing processes, the safety analysis conducted during design and development phase may not be sufficient; consequently, we resolve the deviations from specified behaviours during the operational phase.

Let us consider the autonomous machines operating in Industry 4.0, such as AGVs and robots that are perceived as inter-connected physical objects or things equipped with sensors, actuators and controllers; they are connected via the internet for reasons of data collection and exchange [15]. From the safety perspective, there is a need to determine the potential environments in which they operate, or exposed to, and the movement control measures in certain conditions and areas. As the things have limited power, the cloud and fog services are utilised. The former provides on-demand, broad network access to a shared collection of configurable resources, including storage and processing [16]. The latter extends and brings the cloud platform closer to the things for the resolution of latency, bandwidth and location or context awareness problems, especially for real-time safety-critical functions in Industry 4.0 [17].

An overview of the proposed approach is presented in Fig. 2. The hazard analysis is performed as a first step towards safe operations in Industry 4.0 (see Section 3.1). After that, the safety requirements and safety contracts are derived from the analyses results (see Section 3.2). Next, the construction of safety cases and contracts association is carried out; the safety claims are defined based on the available artefacts, for example, the safety requirements (see Section 3.3). Subsequently, the simulation environments are configured and the operational data is acquired for supporting the data driven production and operations management (see Section 3.4). Finally, to resolve the gaps between the intended and actual behaviour, the safety contracts and safety cases modelled in the OpenCert platform are updated (see Section 3.5). The proposed approach with focus on end-to-end traceability and support of a tool framework can provide a significant boost for the designers to avoid the culture of paper safety at the expense of actual system safety [18].

3.1. Hazard analysis

The factories are often classified as safety-critical, for example, due to simultaneous presence of AGVs, other machines and human workers at the site. The enhanced automation and digitalisation in Industry 4.0 significantly increases the safety issues. To identify hazards, their effects, and causal factors, the hazard analysis is performed. It is the key to prevent or mitigate mishaps. In the context of Industry 4.0, a failure may not just lead to a hazard and accident of a thing/AGV itself. But it can propagate to other things, which lead to a mishap. This is because the things have interactions with one another, and to the fog controller that, in-turn, interact with the cloud infrastructure. In this paper, the hazard analysis is performed by using HAZOP and FTA techniques. The HAZOP analysis is performed to identify possible deviations, operational concerns and their possible fault root causes and consequences. To perform the HAZOP analysis, a set of guide words (e.g., early/late, slower/faster, part of, other than, reverse, omission, before/after, etc.), parameters (e.g., position, distance, detect, speed, etc.), system inputs and outputs, a list of messages and their flow are determined. For in-depth analysis, the fault trees are constructed

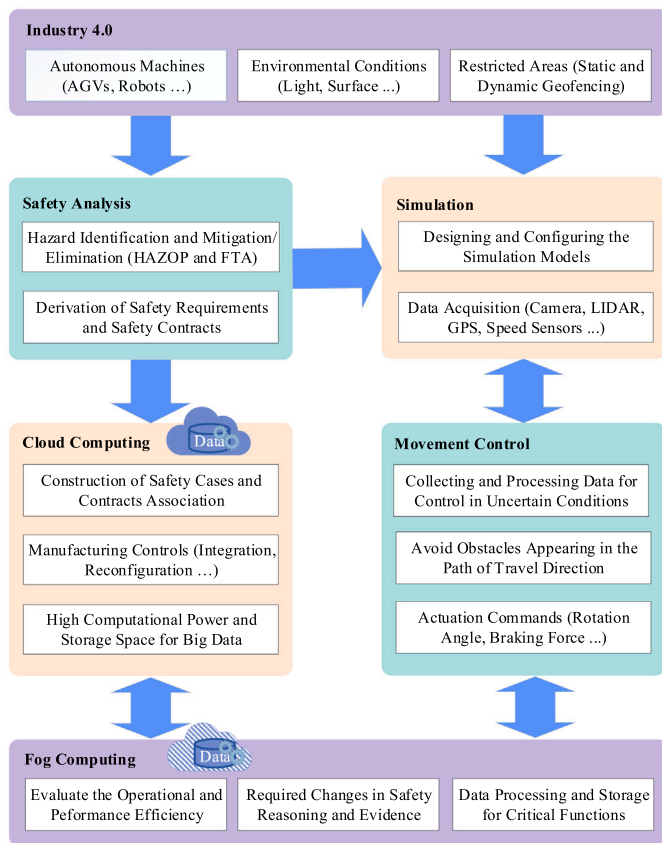


Fig. 2. An overview of the proposed approach.

based on the hazards and their potential effects understood from the HAZOP analysis, in which the identified hazards served as the top-level undesired events [19]. After the identification of hazards, their risks are assessed, and mitigation methods are established that are eventually translated into safety requirements to eliminate or mitigate them.

3.2. Derivation of safety requirements and safety contracts

Safety requirements intend to eliminate or mitigate the identified hazards. Stringent safety requirements are either derived from the hazard analysis or extracted from the safety standards. For example, 'the AGV shall detect and avoid both static and dynamic obstacles appearing in the path of travel direction' is derived from the driverless, automated guided industrial vehicles ANSI/ITSDF B56.5 [20] standard. Industry 4.0 comprises of constituent things and infrastructure elements that are manufactured and provided by different organisations. The manufacturers of the constituent systems, such as cameras and LIDARs (Light Detection and Ranging) are supposed to provide the certification that their products are in compliance with relevant standards (e.g., IEC 61508 [21], ISO 13849-1 [22], and IEC 61496-1 [23]) for the application at hand. If the product is integrated according to the provided instructions, then we can state with confidence that the system meets the safety requirements. The derived safety requirements are implemented in digital twin, and are expected to mitigate the hazards and reduce the risk to an acceptable level [24].

The results gained from the hazard analysis are also utilised to derive the safety contracts that can be further supported by evidence to construct the assurance (safety) cases. The safety contracts demonstrate that the utilised properties of things or infrastructure are sufficient to satisfy the safety requirements and identified failure behaviours can be reduced by a variety of mechanisms, such as testing, monitoring

and operational constraints etc. Since different stakeholders can be involved in Industry 4.0, the information about the assurance of individual things or infrastructure is required in order to derive the safety contracts. However, to be able to provide the assurance, each safety requirement needs to be satisfied by at least one safety contract, and each contract can be supported by one or more evidence(s). The objective of derivation of safety contracts is to associate these contracts with the safety cases as reference points to support the dynamic safety assurance of Industry 4.0. In particular, safety contracts help in determining the validity of safety case modules and overall safety case by comparing safety requirement with related operational data when things or infrastructure are altered or substituted in the system, for instance, in response to changes in market, environmental conditions and device failures. If no deviations are found during monitoring and comparing of data then the assumptions are satisfied in a specific context, and thus, its guarantees also hold. The safety cases and safety contracts also need to be updated with regard to uncertainty aspects.

3.3. Construction of safety cases and association of contracts

The underlying idea of the assurance of Industry 4.0 is to construct the safety cases; firstly, the safety claims are defined, for example, all identified hazards have been eliminated or sufficiently mitigated. Subsequently, the assurance evidence to support the claims is systematically identified, selected, and linked, such that risk is reduced to an acceptable level. In a similar way to the safety-critical applications, to create the safety cases for Industry 4.0, the root claim is broken down into sub-claims by using argument reasoning. However, the modular, dynamic and reconfigurable nature of Industry 4.0, and its architectural levels of things, fog and cloud are specifically considered. The procedure of breaking down claims to sub-claims is continued until we reached claims that can be connected to solutions for which either a direct evidence is specified or an away goal is created by pointing to an already developed module like LIDAR system. The evidence can be obtained during design and development phase from non-formal sources, such as hazard analysis and testing; or from formal verification, such as the results of model checking [25]. In case of Industry 4.0, there is a need to collect the assurance evidence during the operational phase. For instance, the evidence concerning the undeveloped claims requiring the fog processing time could be obtained. The safety case provides the assurance of potential failure behaviour of things or infrastructure, the associated safety contracts help to maintain and update the safety case based on the incorrect or implicit assumptions and deviations from specified behaviours. The safety case (argumentation model and diagram) are constructed in the assurance editor of OpenCert platform, the safety contracts in the form of assumptions and guarantees are specified in an argument element property field.

3.4. Data driven production and operations management

IoT data will be generated and renewed in the Industry 4.0 at a very high speed. Multiple types of sensors such as LIDAR, Camera, Global Positioning System (GPS) are used for supporting the data driven production and operations management. They are either placed on top of things (such as robots and AGVs) or mounted on appropriate locations in the environment (such as on the walls, pillars), for covering specific areas. Let us consider the scenario of transportation and distribution of materials for which AGVs are used. To establish a basis for evaluation and improvement of performed operations, an appropriate simulation environments needs to be configured. In our demonstrations we used Carla [26], AirSim [27], Webots [28] and Oryx.³ The mounted cameras provide support for detecting and recognising obstacles appearing in the path of travel direction; message signs and lane markings can

³ <https://www.oryx.se/>.

also be considered. However, the LIDAR provides better support for distance measurements and dealing with environmental conditions. Besides the cameras and LIDAR, the GPS would be fitted for tracking and navigation purposes. They transmit data at a rate of 20–60 MB/s, 10–70 MB/s and 50 KB/s, respectively.

In the Industry 4.0, the production operations, for instance, automated loading is carried out in a collaborative manner. This means that the AGV operates in conjunction with other AGVs and machines. The site management is perceived as a fog controller. It gathers the data from different machines and is responsible for managing operations including the mission termination or assignment to the AGVs, and traffic control to avoid collisions. The communication between the AGVs and fog controller also supports entry control, i.e., to permit or restrict entry to geofenced areas. The data produced by sensors placed on specific things and areas is processed by the fog controller. A detailed list of parameters is established to support the data driven production and operations management. This step is fundamental to evaluate the operational and performance efficiency, and to deal with the deviations between the intended and actual behaviours at the operational phase.

The modelling of safety cases and association of contracts is carried out in the OpenCert platform. The safety cases can be stored in the workspace directory, or in the CDO, which is both a development-time model repository and a run-time persistence framework. The repository sessions provide support for obtaining and modifying them. The features provided by CDO satisfy the design- and run-time requirements of the self-adaptive cloud applications [29]. In this sense, the fog controller is not just connected to things and remote resources, but also serve as a bridge between them. The work presented in Seybold et al. [29] discuss the experiences with EMF and CDO in scalable cloud scenarios. The in-browser Eclipse IDE for cloud (Eclipse Che⁴) may also be used. In addition to the safety cases, the cloud level is considered for some manufacturing controls and high computational power and storage space for big data except for real-time safety-critical functions in Industry 4.0.

3.5. Identification and resolution of gaps in safety cases

There is a need for the identification and resolution of gaps between the intended behaviour reflected in safety cases and the actual safety of production operations in Industry 4.0. For this reason, the operational data is utilised. In particular, the safety contracts constructed for uncertainty sources are monitored based on the gathered data. As mentioned in Section 3.3, the safety contracts describe the functional and behavioural properties in the form of assumptions and guarantees. The safety case modelled in the OpenCert platform is traversed to retrieve the elements IDs and associated safety contracts; this information is stored in the data structures. In the safety cases, the contracts are associated in the argument element property field, named as content. To check the validity, the safety contracts are compared against the list of parameter values obtained during the operational phase that includes the distance to the obstacles, location, path points, weight, speed limits, and time frame for transmission, data processing and actuation of given commands. This provide the means to cope with the completeness, consistency and correctness problems.

Based on the comparative analysis of operational data with the assumptions made and predicted guarantees, the evidences are obtained, which are attached to the undeveloped claims, located with their element IDs. Otherwise, the deviations between the intended and actual behaviour are tracked. Let us consider the processing and actuation times, if they are increased, the safe distances to avoid the obstacles is increased, or the speed is reduced in alignment with the determined threshold. The positive impacts of advanced technologies are also considered; they could not only increase the detection capabilities, but

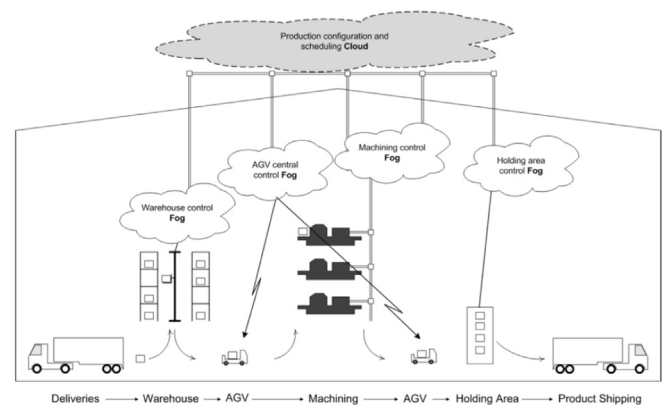


Fig. 3. Materials transportation and data flow in industry 4.0.

also decrease the transmission and processing time frame. In this case, we explicitly evaluate the corresponding impacts on the control actions. The confidence, in certain situations, is gradually increased to resolve the discrepancies. The update of safety contracts and safety cases is carried out based on the optimal actions. The required changes in safety cases are tracked on the fog level and then the update command is issued. The safety contracts and safety cases are updated on the CDO server, which is accessed by the OpenCert argumentation editor connected to the CDO server.

4. Case study

To transform the raw materials into the final products, the evolution of traditional supply chain is considered critical for industry 4.0 [30]. For the enhancement of automation and digitisation of supply chain, the alignment with interoperability, information transparency, technical assistance and decentralised decisions is suggested [31]. Let us consider a synthetic but realistic and generic use case that concerns transportation and data flow in the Industry 4.0. The use case chosen in the FiC project⁵ and AAIP [32] consists of a number of computer-based machine tools that interacts with each other for product manufacturing. There are few manufacturing controls that are located remotely on the cloud, whereas the others are managed locally at the fog or things levels. The computer-based machine tools make a range of gearbox shafts from metal blanks. The weight of a blank is about 4 kg. They are delivered in pallets of 50 and stored in an automated warehouse until they are required. The finished products are also packed into pallets and taken to a holding area before being shipped to the main assembly plant. The transportation and distribution of pallets are performed with battery-powered AGVs. Fig. 3 shows the conceptual flow of materials and data in the Industry 4.0.

The AGVs are widely used in industries. Their operations are similar to the autonomous haulers that are used for rocks transportation in the advanced quarry site [19]. The AGVs shall be equipped with device(s) to detect and avoid obstacles appearing in the path of travel direction [20]. To carry out the transportation and distribution of materials efficiently, the next generation of AGVs, especially for Industry 4.0 could be equipped with autonomous driving technologies. The cameras and LIDARs provide means for emergent interactions with machines, operating speed increase, and safety of AGVs; they can be mounted on top of the AGVs or otherwise infrastructure level for the detection and recognition of obstacles appearing in the path of travel direction. Since the LIDARs are relatively expensive, if they are mounted on the infrastructure level, then the limited number of LIDARs can fulfil the

⁴ <https://www.eclipse.org/che/>.

⁵ <http://www.es.mdh.se/fic/>.

needs. This is also relevant in situations of failure of obstacle detection devices mounted on top of the things, for instance, the transition is made to other devices that are placed in hazardous zones like loading and unloading. Accordingly, besides the command/control functions, the ways for transmitting, processing and storage of big data are mapped to the generic architecture of Industry 4.0.

4.1. Hazard analysis

The performed analysis not just focuses on the individual behaviour of AGV, but also the emergent interactions of AGVs, with fog/cloud server or other working equipment. The identified hazards can be categorised into: *sub-system failure hazards* occurred due to unavailability, unreliability or incorrectly structured software/hardware component, such as AGV onboard system failure and brake system failure; *deviations from procedures*, for example, incorrect application of LIDAR or inadequate installation of software; *communication failure*, for instance, a message is received too late, an incorrect message is transferred, wrongly interpreted by the receiver, or network unavailability; *operational hazards* such as failure of obstacle detection and collision avoidance mechanism; and *environmental hazards* which involve extreme heat or cold, light, and driving on rigid surfaces. Once hazards are identified and classified, the corrective measures are drawn to eliminate or reduce the risk of the identified hazards to an acceptable level.

The results obtained from the HAZOP and FTA demonstrate that critical incidents can occur in Industry 4.0, if correctly and timely communication is not established. For instance, AGV received a message too late than expected from fog controller concerning position and distance to obstacle. The reception of late message can be caused by AGV fog failure, downtime, or network unavailability, which may lead to the incomplete mission, a collision of AGV with dynamic objects (e.g., other AGVs, human worker, or other working equipment), or with static objects (e.g., ladder, fallen pallet). These collisions can be catastrophic, especially if the AGV carries dangerous materials (e.g., explosive, toxic, etc.). This can be prevented by increasing number of wireless access point or getting the information from another fog. Moreover, the failure of obstacle detection and collision avoidance mechanism (caused by LIDAR or camera failure) may also lead to a collision of AGV. This can be prevented by getting the information from the sensors mounted on the infrastructure level. In this paper, we focus on the hazard like collision of AGV with static objects and dynamic obstacles. It is assumed that if collisions happen while AGV is stopped, then it must be the moving obstacle's fault.

4.2. Derivation of safety requirements

The risk reduction and mitigation measures identified through the hazard analysis are translated into the safety requirements. Additionally, safety requirements are extracted from the automated guided industrial vehicles ANSI/ITSDF B56.5 [20] standard. To address industry-level hazards different safety requirements specified, such as "The AGV shall not collide with obstacles appearing in the path of travel direction" is considered as a main requirement. To achieve this requirement, several other requirements have been specified. Some of them are applicable at the system level, while others are relevant for the software and hardware functions.

SR 1: The AGV shall detect static and dynamic obstacles appearing in the path of travel direction.

SR 1.1: The obstacle and position detection devices shall be sufficiently robust and works in practice.

SR 1.2: The obstacle detection device mounted on AGV shall cover a minimum range of 10 m.

SR 1.3: The obstacle detection device shall detect indoor obstacles with the worst accuracy of $\pm 6\text{cm}$.

SR 1.4: The GPS fitted on the AGV shall estimate position with the worst accuracy of $\pm 30\text{cm}$.

SR 2: The retrieved data shall be processed to determine risks and preventive measures within maximum 800 ms.

SR 2.1: The transmission of data retrieved from object detection and positioning devices to the Fog controller shall take maximum 50 ms.

SR 2.2: The Fog controller parse the data to detect moving direction, distance and current speed of the obstacles that shall take maximum 500 ms.

SR 2.3: The Fog controller shall send the parsed data to the AGV within maximum 50 ms.

SR 2.4: The AGV's onboard system shall determine protection measures within maximum 200 ms.

SR 3: The AGV shall avoid obstacles for which deceleration and steering rotation are commanded.

SR 3.1: The actuation time for deceleration and steering rotation shall not be greater than 100 ms.

SR 3.2: If static obstacle is detected in travel path, the maximum speed limit 1.2m/s shall be maintained in hazard zone until the remaining distance of 2 m.

SR 3.3: If dynamic obstacle is detected in travel path or junction, the maximum speed limit 0.3m/s shall be maintained in critical zone until the remaining distance of 2 m.

SR 3.4: The safe distance to the obstacles after stopping shall not be shorter than 1 m.

SR 3.5: If obstacle is detected, the AGV outbreak from the original path to avoid the obstacle for which the smallest deviation from the original path should be chosen as side preference.

4.3. Derivation of safety contracts

After identifying the safety requirements, safety contracts are documented for uncertainty sources. Let us assume that the failure of LIDAR or speed sensor, or transmission delay can be hazardous, then the corresponding safety requirement is addressed by a contract assumption, whereas guarantee shows that the failure is appropriately handled by making the transmission to another detection device, to map detecting distance covered within timeframe and speed is always limited to $\leq 0.3\text{m/s}$. Table 1 shows a template of derived safety contracts, where A and G stand for Assumption and Guarantee, respectively.

4.4. Development of safety cases using contracts

Once safety requirements and safety contracts are derived for Industry 4.0, the safety case is created, for which the safety requirement "The AGV shall not collide with obstacles appearing in the path of travel direction" satisfied to SIL 2, is selected as a root claim, as shown in Fig. 4. This claim is decomposed into two sub-claims using a strategy "Arguments over element contracts". In particular, the claim about completeness and correctness of hazards identification to maintain safe distance of AGV from other AGVs, working equipment and human workers, and the claim about the detection and avoidance of static and dynamic obstacles are created in the context of things and infrastructure elements. This procedure is continued until there are elementary claims that can be connected to the available evidence (i.e., InformationElementCitation Property type = "solution"). An away goal (i.e., ArgumentElementCitation Property type = "claim") is created, for the claim "The transmission of data between fog controller and AGV shall take maximum 50 ms", and "Argumentation Reference" property indicates the reference to the module in which it is developed (i.e., Infrastructure module). The supplier of fog services has provided

Table 1
Template of derived safety contracts.

A1:	For the operating AGV, no obstacle detection device failure AND no mechanical parts failure AND no transmission delay AND no obstacles detected within 10 m range of travel path.
G1:	The higher speed of AGV is maintained.
A2:	For the operating AGV, obstacle detected in 10 m range of travel path AND the obstacle is not moving (i.e., static obstacle is detected).
G2:	Until the 2 m distance, the speed is always ≤ 1.2 m/s and the smallest deviation with side clearance ≥ 0.5 m is chosen to circumvent the obstacle.
A3:	For the operating AGV, obstacle detected in 10 m range of travel path AND the obstacle is moving (i.e., dynamic obstacle is detected).
G3:	Until the 3 m distance to the dynamic obstacle, the speed is always ≤ 0.3 m/s and the AGV stops before the remaining distance 2 m.
A4:	The LIDAR is mounted on top of AGV or infrastructure as per the manufacturer recommendations.
G4:	The LIDAR shall detect indoor obstacles within the range of 10 m with the accuracy of ± 6 cm.
A5:	For the operating AGV, LIDAR sensor failure OR wheel speed sensor failure OR transmission delay.
G5:	The transition is made to another detection device, to map for detecting distance covered in timeframe and the speed is always ≤ 0.3 m/s.
A6:	AGV Fog controller is independent AND accurate AND distinct input data is received within 50 ms timeframe.
G6:	AGV Fog controller accurately computes the received inputs and provides data in 500 ms time frame.

the assurance evidences for it. The claims, such as “The fog controller parse the data to detect moving direction, distance and current speed of the obstacles that shall take maximum 500ms” are created as undeveloped claim by selecting the property toBeSupported = “true” means that it requires further development. The Assumption and the Guarantee properties for the contract are specified in the Content field of related claims in assurance case editor. For the undeveloped claim, the contract allows that its guarantees, can be supported by artefacts (e.g. the latter referring some verification results, simulation runs).

The claim about “The AGV provides the required properties for handling collisions with static and dynamic obstacles” is decomposed into sub-claims about providing sufficient properties or failures behaviour of mechanical parts, position and obstacles detection devices, and AGV on board systems using three strategies. The mechanical parts of AGV are assumed to be well maintained, and have not any failure, which may not always be the case. Therefore, the claims about mechanical hazards identification and mitigation can be supported by hazard analysis results (e.g., mitigation measures taken to eliminate or prevent the failures), inspection and/or testing reports. The suppliers of camera, LIDAR and GPS are able to provide assurance for their products that support various claims about camera, LIDAR and GPS as detailed in the safety contract. Therefore, we used away goals for them such that these claims are developed in different modules. As we can see in Fig. 4, it is assumed that the LIDAR mounted on AGV will provide point clouds in 10 m range with the worst accuracy of ± 6 cm. The confidence in this claim is provided by the LIDAR safety case.

4.5. Transportation and data flow in industry 4.0

In the traditional AGVs, besides the infrared sensors that are mounted to detect obstacles, the camera is placed to give a clue for the lines drawn on the industry floor. This behaviour is analysed with the e-puck robot, which is configured in the Webots simulator. The Radio Frequency Identification (RFID) tags are also used to locate the AGVs. The data produced by eight infrared sensors, camera with limited resolution (52×39 pixels), or RFID tags can be easily handled by the AGVs that are perceived as things. In contrast to this, a huge volume of data could be transmitted to the fog level; it acts as a bridge

between things and cloud computing levels. The different formats of data from diverse devices for instance high-resolution cameras and LIDAR is gathered from configured simulators. The acquisition and integration of data can be carried out at things (AGVs) level. The fog level distinguishes the real-time safety-critical data for processing and storage itself. Otherwise, it further transmits the data to the cloud. In this sense, the fog improves the computation and communication efficiency by reducing the quantity of data to be transmitted to the cloud. On the other hand, the fog also forwards necessary commands to thing to fulfil the specific production demands. Since the considerable data, including safety cases is stored in the cloud, of course, the ways for retrieval and update are important.

4.6. Dynamic update of safety contracts and safety cases

In addition to the risk control actions, to avoid the culture of paper safety, the update of safety contract and safety case is performed. The safety contracts are associated with the safety case elements, which are retrieved and monitored. The gaps with the contracts, i.e., the assumptions made and guarantees provided indicate that the safety contract is broken. As discussed in Section 3.5, the deviations or mismatches are tracked by comparing the safety contracts with their respective parameters that are gathered from the simulations. This provides a basis for the adaptation of affected parts of safety cases and associated contracts. If the failures, delays and obstacles are not detected, i.e., the assumption “for the operating AGV, no obstacle detection device failure AND no mechanical parts failure AND no transmission delay AND no obstacles detected within 10 m range of travel path” is satisfied. Then the “higher speed of AGV is maintained”. The speed limit 14 km/h is reached. In case the assumption “obstacle detected in 10 m range of travel path AND the obstacle is not moving” is satisfied, the predicted guarantee hold “until the 2 m distance, the speed is always ≤ 1.2 m/s and the smallest deviation with side clearance ≥ 0.5 m is chosen to circumvent the obstacle”. In this way, simulation result related to the satisfied safety contracts is assembled as an evidence to the undeveloped claim “propsAvoidSat”.

The assumption A5 “for the operating AGV, LIDAR sensor failure OR wheel speed sensor failure OR transmission delay” is checked. If the

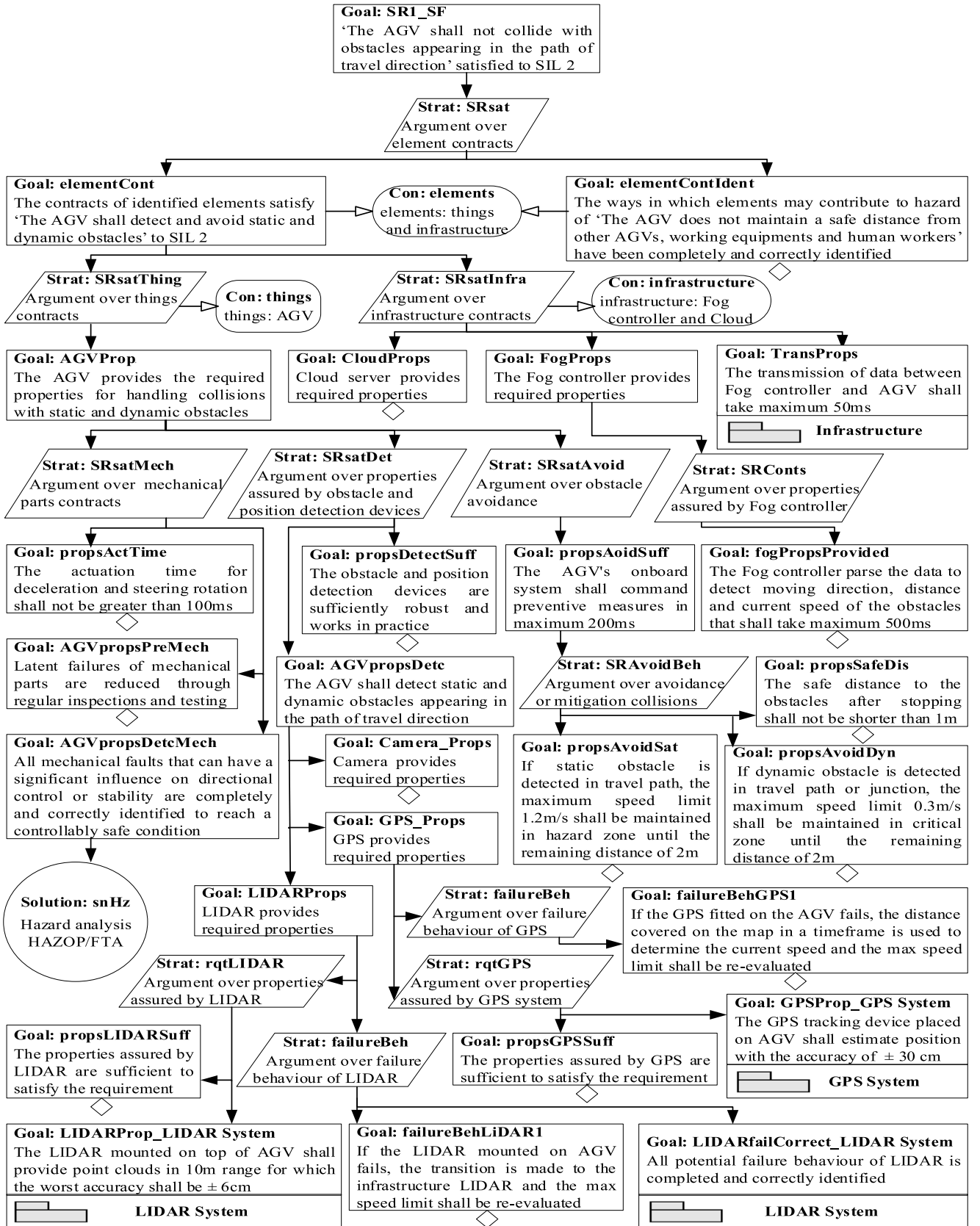


Fig. 4. A safety case for the considered Smart Factory.

LIDAR sensor mounted on AGV is failed then “the transition is made to another detection device”. Such devices are placed on the specific zones. To reduce risk to an acceptable level, the speed limit at first

stage is set to ≤ 0.3 m/s. The “transition is made” to the infrastructure devices, or the other AGV is followed in terms of platoon to accomplish the intended operation. The confidence in terms of the operating speed

is gradually increased. In case of speed sensor failure, the maps are used for “detecting distance covered in timeframe and the speed”. The additional safety countermeasure have also been considered for the AGVs with faulty (sub)system. Depending on the severity risk factor, the other AGVs in close ranges can be commanded, e.g., to drive away. The safety contract linked to the claim “fogPropsProvided” guarantee that the “AGV fog controller accurately computes the received inputs and provides data in 500 ms time frame”. This, however, is dependent on the conditions “AGV fog controller is independent AND accurate AND distinct input data is received within 50 ms timeframe”. Due to the transmission delay, the contract assumption A6 does not satisfy and the AGV fog controller may not hold the predicted guarantee G6. As a consequence, the AGV received the command late than expected and therefore corresponding control actions need to be evaluated. In particular, the threshold is determined, slow down and stopping time would be accordingly increased and updated. The alternative to cover the threshold of 200 ms is to reduce the maximum speed limit of AGV to 12 km/h. Besides the performance degradation, the upgradation is taken into consideration. The adaptations in safety contracts and corresponding safety argument elements are triggered in response to the selected control actions. In contrast to the matching of parameter names and their values/ranges for safety contracts, the text-based matching is performed to alter the description of safety case elements.

5. Related work

Denney et al. [4] discusses the dynamic safety cases as an operationalisation of the concept of through-life safety assurance; the data produced from safety management system can be utilised to create continuously evolving assurance argument. The suggested life cycle of dynamic safety cases comprises four main activities: identify the sources of uncertainty that weaken the confidence; monitor the deficits in safety argument(s) by collecting operational data; analyse operational data for the determination of threshold defined for assurance deficits; and respond to the operational events that affect safety assurance. This concept highlights a number of challenges, for example, how to decide the most important subset of uncertainty sources and how to automate them. The research in McDermid et al. [5] proposed the framework for safety assurance of automated systems that also has four major elements: the ‘real world’ consists of automated system and operational environment; the ‘world as imagined’ contains models (design and simulation) and results of safety analysis achieved based on them; the ‘world as observed’ involves the collection and processing of operational data; and the ‘safety case’ that previously reflects the ‘world as imagined’ but then updated to respond to the reality ‘world as observed’. Jaradat and Punnekkat [6] describes the monitoring of runtime failures related to the hardware component and failures analysis by comparing with a predefined threshold. The fault trees were used for deriving safety contracts and defining thresholds.

Sljivo et al. [33] extends the notion of contract with the strong and weak contracts to handle the multi-context setting of reusable components by distinguishing assumptions on configuration parameters and operational variables. The strong contracts captures properties in all contexts and must be satisfied; whereas the weak contracts not need to be satisfied. Similar to the product line, the configuration aware contracts can be used to assign the minimum requirements. The boolean parameters are defined to capture behaviours in terms of contracts; therefore, a lot of configuration parameters for different contracts are needed. In another study, Sljivo et al. [34] discussed the tool-support for specifying the contracts and argument-fragments generation. Besides the informal description of each of the contracts, the relations to the requirements are added and the status update is performed to proceed with the argument-fragments generation for individual components in the system.

Calinescu et al. [7] present ENTRUST methodology for the engineering of trustworthy self-adaptive software systems: unmanned

underwater vehicle and foreign exchange. ENTRUST supports the partial instantiation of argument pattern at design phase, and adaptation of evidence produced by the monitor-analyse-plan-execute (MAPE) control loop to fill in the placeholders from partial assurance argument. For modelling and verification of MAPE models, UPPAAL and PRISM model checkers are utilised. Denney and Pai [35] present a methodology that combines the manually created, aircraft-level argument fragments derived from safety analysis, with an automatically generated lower-level argument fragments derived from formal verification. The formal verification is supported with the AutoCert tool, while the AdvoCATE toolset is utilised to assemble both of the fragments created for Swift Unmanned Aircraft System (UAS). However, the evidence of safe flight and other evidence from operational data, that is operational safety cases are not considered. In another paper, Denney and Pai [36] combines safety assurance arguments with bow-tie diagrams as a common framework to support modelling of UAS safety cases for airworthiness and operational safety. The abstract safety architecture specifying the collection of hazard controls is represented in bow-tie diagrams and an argument architecture is presented as argumentation patterns.

For the systems with enhanced automation and digitalisation, the safety analysis conducted during design and development phase may not be sufficient. The dynamic safety assurance is perceived as fundamental for them [4,5]. The research in [2] highlight the safety assurance challenges for Industry 4.0. This paper, however, is a step towards explicitly supporting the dynamic safety assurance of Industry 4.0. After the safety analysis conducted during design and development phase, the digital twins are configured, the operational data is gathered from them to monitor, evaluate, and resolve deviations from the specified behaviour reflected in safety cases and the actual behaviour.

6. Conclusions and future work

This paper targets the construction and maintenance of safety cases for Industry 4.0. The hazard analysis techniques, particularly HAZOP and FTA, are applied to identify unplanned events or hazardous conditions, their effects on the behaviours and the reliability of transportation and data flow in Industry 4.0. The results gained from the hazard analysis are utilised to derive the safety requirements related to the prevention or mitigation of potential hazards. Safety contracts that support the maintenance and adaptation of safety cases based on operational deviations, incorrect or implicit assumptions and guarantees have also been derived. The OpenCert platform is utilised for safety case construction and association of derived safety contracts with the safety case. To establish a basis for evaluation and improvement of performed operations, simulation environments have been configured. The operational data is gathered from them to monitor the uncertainty sources, track and evaluate deviations with the intended behaviour, and resolve the gaps in safety cases to reflect the actual behaviour of production system. The obstacle detection, data processing and avoidance aspects are specifically taken into consideration. The usefulness is demonstrated for the materials transportation and data flow in Industry 4.0.

In further evaluation, we plan to explore other type of machines that have different characteristics and transformation of application domains to the Industry 4.0. The dynamic safety assurance methodology is generally applicable to various automated machines. The initial effort is although required to derive the safety contracts for uncertainty sources, but it is worthwhile to detect deviations from intended behaviour and perform necessary adaptations at the operational phase. In this process, there is a need to handle the short- and long-term deviations in different ways. The dynamic risk management is crucial for short terms deviations, but the safety contracts and safety cases may just be updated for the frequent or long term deviations.

In another work, the virtual boundaries around geographic zones (i.e., geofences) are used as a measure for elimination or mitigation of

automated transportation risks at the operational phase [24]. Specifically, the static, dynamic, time-based and conditional geofences are taken into consideration. In the current circumstances when the different configurations are reflected in the safety cases, problems arise, due to unavailability of operational data for all the parts. Accordingly, we plan to support the safety case configurations as part of the dynamic assurance in Industry 4.0. The configuration analytics will be performed to determine the production safety and operational performance, as well as the tradeoffs between them. Another limitation is that the identification and resolution of gaps in safety cases is dependent on the GSN and the OpenCert platform. In the future, the safety cases modelled in other formats, such as free text and tabular structures could be supported.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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